

1. Write a Java POJO (plain old java object) named "Employee", inside Employee class, you should have

1. Employee's name
2. Employee's Date of Birth
3. Employee's Department (it can be another POJO)
4. Employee's Social Security Number
5. Employee's home address (it can be another POJO)

Override toString method, so that when an Employee object is being printed, the print out looks meaningful and readable.

Override equals method, so that only when two Employees have identical information, we consider they are the same employee.

2. Write code to instantiate at least two instances of above Employee class, use code snippets to explain how these Employee objects are allocated to JVM memory. You may use java reflection utilities to demonstrate it.
3. Write static utilities in your Employee class, demonstrate how static content differs from others during class instantiation.
4. Explain why global variables are NOT recommended, you may use code snippets.
5. Explain why Strings in Java are considered "Immutable"?
6. Write code snippets to explain what does "Final" keyword do, and what we need it?
7. Write code snippets to explain why Java is "pass-by-value", and why do some people think it might be "pass-by-reference"?
8. Write code snippets to explain overloading in Java, explain how does Java define method signature.
9. You may be asked to demo and explain your answers/code in class.
10. Use Java collection framework datastructures (e.g. Set, Map, List) to solve following Leetcode questions, you MUST use Java and you MUST use datastructures provided by Java Collection framework:
 1. Top K Frequent Elements (LeetCode 347)
 2. Two Sum (LeetCode 1)

This is a separate coding task other than daily leetcoding questions, please submit your solutions as part of this assignment, rather than in the leetcoding sheet.

11. You can either have your code snippets in your markdown file, or include in a Java project.